Homework #1 for E1.216 Computer Vision (2026)

Notes

- You may not discuss any aspect of this homework with others. Do it on your own.
- Submit the entire homework as a hardcopy in class on 4 March 2026. **No extensions.**
- Please save your code. I may ask to see it.

Question 1 [10 points]

Prove that an affine transformation on $\mathbb{R}^2$ preserves parallelism.

Question 2 [10 points]

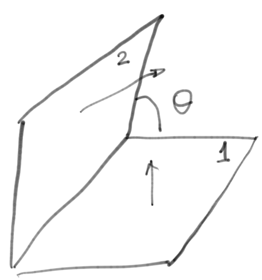

Figure 1: Two plane configuration for Question 2

Let us consider two Lambertian planes of a constant albedo ($\rho = 1$) as shown in Figure 1. Let the angle between the two planes be θ. We use a point-source-at-infinity light source to illuminate the scene. Under a given lighting direction l_k, let the observed intensities for planes 1 and 2 be I_{1k} and I_{2k} respectively. Develop a solution for estimating the angle between the planes θ when we are given measurements for 3 different lighting directions, i.e. $k = \{1, 2, 3\}$. [10 points]

Question 3 [30 points]

- Consider points $\mathbf{p} = [x, y, 1]^T$ in the 2-D plane. Derive the equation for a circle in the form $\mathbf{p}^T\mathbf{C}\mathbf{p} = \mathbf{0}$. **Hint:** $\mathbf{C}$ is a symmetric matrix.
- Further consider the case where the observed points have noise in their locations and some points are outliers. Briefly develop the RANSAC and M-estimator (IRLS) for estimating the parameters of the circle.
- Implement and test both approaches. Give me your general observations from this exercise.